JEEWANTHA LAHIRU

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PROFILE

A passionate Senior Software Engineer currently pursuing an MSc in Computer Science at the University of Peradeniya, with a BSc (Hons) in Software Engineering from the University of Kelaniya and nearly five years of professional experience in game development and software engineering. Currently working as a Senior Software Engineer at SimCentric Technologies, specializing in Unreal Engine development, gameplay systems, simulation technologies, and immersive interactive experiences.

EDUCATION

Master of Science in Computer Science (Ongoing) **Mar 2026 - Present**

University of Peradeniya, Sri Lanka

- Studying advanced areas of Computer Science, including Artificial Intelligence, Machine Learning, Neural Networks, Advanced Computer Graphics, and Computer Vision.
- Exploring Large Language Models (LLMs), AI-driven application development, intelligent agents, and emerging agentic AI technologies.
- Developing knowledge of AI orchestration, workflow automation, and the integration of intelligent systems into modern software applications.
- Exploring deep learning, intelligent computational techniques, and their applications in solving real-world problems.

Bachelor of Science (Honours) in Software Engineering **Feb 2019 - Jul 2023**

University of Kelaniya, Sri Lanka (GPA - 3.45)

- Specialized in Big Data, Distributed and Cloud Computing, Game Development, Mobile Development, and Semantic Web.
- Thesis on "A Driving Simulator based on Serious Gaming Technique to reduce the Occurrence of road".
- Developed an Online Exam Monitoring App using camera-based behavior analysis and machine learning to detect and notify potential cheating, ensuring fair examinations.

WORK EXPERIENCE

Senior Software Engineer, Simcentric Technologies **Aug 2025 - Present**

Colombo, Sri Lanka

- Primarily involved in the development of SAF-TAC, a scalable, low-overhead military training and virtual reality (VR) simulation platform built on Epic Games' Unreal Engine, designed to deliver immersive and realistic defence training experiences.
- Contributed to the development of a multi-platform training environment supporting Desktop and VR, facilitating individual and collective training, mission rehearsal, tactical experimentation, and immersive training scenarios.
- Developed interactive simulation systems covering vehicle mechanics, drone simulation, and character development, implementing gameplay functionality and simulation behaviours using Unreal Engine C++ and Blueprints.
- Implemented data-driven systems and workflows to manage application behaviour, gameplay configurations, and simulation parameters, improving flexibility, maintainability, and reusability across the platform.
- Designed and implemented gameplay and simulation features with a focus on modular architecture, reusable components, realistic interactions, and maintainable code structures.
- Worked on integrating complex gameplay mechanics and simulation systems, ensuring reliable interactions between different components within large-scale virtual training environments.
- Contributed to the RAPTOR project, developing Unreal Engine-based drone simulation integrated with the PX4 flight control system through MAVLink communication.
- Implemented communication between the simulated drone and external flight-control systems, enabling remote drone control and bidirectional exchange of flight telemetry and sensor data.
- Integrated a PX4 instance running on an Ubuntu server to generate and transmit automated flight-control commands and telemetry data to the Unreal Engine simulation.
- Applied strong C++ development practices, object-oriented programming principles, and software engineering best practices to develop efficient, maintainable, and high-performance simulation systems.
- Focused on code optimization, debugging, system integration, and performance improvements to ensure reliable execution across complex real-time simulation environments.

Visiting Lecturer, University of Kelaniya **Mar 2026 - Present**
Dalugama, Kelaniya, Sri Lanka

- Delivered undergraduate lectures for the Software Engineering degree programme, covering:
  - SENG 41323 – Games Design, Artwork, and Programming.
  - SENG 42323 – Advanced Topics in Game Design and Animation.
- Designed and delivered practical learning sessions covering Unreal Engine, C++ programming, game design principles, gameplay systems, artificial intelligence, animation, and character development.
- Introduced students to industry-standard game development workflows, gameplay architecture, and real-world software engineering practices.
- Guide final-year students in developing practical game projects, applying advanced gameplay programming concepts and translating technical requirements into functional solutions.
- Shared industry experience and practical insights to prepare students for professional opportunities in the game development industry.

Game Programmer, G-Devs **Apr 2024 - Aug 2025**
Warszawa, Poland

- Leading the development of Rich Dad Simulator, a first-person retail management simulation game built using Unreal Engine as part of a four-member development team, serving as the lead developer.
- Designed and implemented core gameplay systems for retail business management, including staff management, inventory control, pricing, customer interactions, and store expansion mechanics.
- Developed and integrated character systems, NPC AI, animal behaviours, vehicle AI, and interactive gameplay mechanics to create dynamic and immersive simulation experiences.
- Implemented gameplay features primarily using Unreal Engine Blueprints, developing custom C++ plugins and integrating them into projects whenever advanced functionality or performance requirements demanded it.
- Designed and implemented user interfaces (UI), HUDs, and interactive widgets, focusing on usability and human-computer interaction (HCI).
- Worked on gameplay optimization, debugging, bug fixing, and development testing to improve system stability, performance, and overall gameplay quality.
- Applied software engineering best practices, version control workflows, and collaborative development practices throughout the development lifecycle.

Associate Software Engineer in Game Development - Arimac **Jan 2022 - Apr 2024**
Colombo, Sri Lanka

- Two years of professional experience as an Associate Software Engineer, specializing in Unreal Engine game development.
- Managed the Epic Games Marketplace team, coordinating task assignments, overseeing asset development, and managing product publishing.
- Contributed to the development of NERO, focusing on gameplay programming, feature implementation, and functional testing.
- Coordinated development workflows to maintain asset quality and ensure timely project delivery.

Intern Software Engineer in Game Development - Arimac **Jul 2021 - Jan 2022**
Colombo, Sri Lanka

- Completed a professional internship in game development, gaining hands-on experience in Unreal Engine and developing foundational skills in gameplay programming, Blueprint scripting, and interactive game systems.
- Assisted in implementing gameplay features and exploring core game development workflows using Unreal Engine Blueprints.

ADDITIONAL INFORMATION

- **Technical Skills:** Unreal Engine (C++ & Blueprints), GitHub, Bitbucket, Perforce, Jira, Agile Software Development
- **AI Tools & Technologies:** Claude (Opus) with code integration, ChatGPT Go
- **Languages:** English, Sinhala
- **Awards/Activities:**
  - Contributor – ACES Coders 8.0 Hackathon, University of Peradeniya.
  - Vice President – Software Engineering Students' Association, University of Kelaniya.
  - Member – IEEE Student Branch, University of Kelaniya.
  - Participant – IEEE Django Workshop.
  - Guest Speaker – EvolveX 2025 (Game Development & Esports), University of Kelaniya.
  - Interviewer – Mock Interview Series, University of Kelaniya.
  - Guest Speaker – Software Engineering Career Pathways, Sabaragamuwa University.
  - Guest Speaker – FutureStack 2026, University of Kelaniya (Digital Gaming and Animation Applications).
  - Podcast Guest – Gaming Hackathons, University of Kelaniya.